Cluster Quasimodels for $\mathcal{ALCI}_{\mathrm{RCC5}}$: a Round-15 Decision Layer without Automata

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Abstract

The ninth review certified the semantic half of the round-14 stack (the active-virtual-bag amalgamation and truth lemmas) and located the open problem precisely: the abstraction's relational enrichment lives off any finite tape, so no finite tree automaton was ever shown — and, as sketched, none soundly can be shown — to decide its existence. This manuscript removes the automaton layer entirely. The decision object is a *cluster quasimodel*: a finite set of Hintikka types, the safe atomic links between them, and a finite catalogue of small closed atomic *cluster patterns* in which every existential demand is fulfilled within one step, together with finitely checkable long-range conditions formulated over the *fold lattice* — the closure of the RCC5 atoms under set-composition, which is shown (machine-checked) to contain exactly ten sets, every one of which is a vertical singleton or contains a horizontal atom. Global coherence is not enforced by a decision device at run time; it is *constructed once, in the soundness proof*, by iterated patchwork over the unfolding tree together with a canonical horizontal selection on the fold lattice — the discipline that survived reviews eight and nine as the Shadow Amalgamation Lemma, now applied at the level where it belongs. Eventualities need no parity condition: cluster patterns forbid promissory witnesses, so fulfilment is one-step by construction. Uniquely realizable types (the strong-equality towers of the ninth review's probe p4) are handled by a shared core cluster. Decidability follows from the finiteness of the certificate space and the finite checkability of its conditions; the only remaining external input is the RCC5 patchwork property — the two-way-automaton emptiness dependency is gone. The width bound is re-proved under a single decreasing measure, resolving the ninth review's F3. Status: the decidability theorem remains *strongly supported, not certified.*

Contents

*Round-15 manuscript, responding to the ninth review's recommendations R1–R6 (the cold review of the round-14 stack, `papers/cold_review_round14/`): it takes the review's alternative R2 — replace the tree-automaton layer by a mosaic-style argument executed on the infinitary abstraction — and supplies the finite certificate notion, its soundness construction, and its extraction. Project conventions apply: nine reviews so far, nine defects; this manuscript is unreviewed and should be presumed to contain one. Its designated weak points are stated in Section 10.

1 Position of this manuscript, and why no automaton

1.1 What is inherited, what is replaced

Inherited, as certified or repeatedly-found-sound components:

- the **semantics**: abstract complete atomic RCC5 frames (Δ, ρ), $\rho : \Delta^2 \to \mathsf{Rel}$, $\mathsf{Rel} = \{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}, \mathsf{DR}\}$, with identity (R1), converse (R2), composition (R3) for the standard table, strong equality; concepts in NNF; closure $\mathrm{Cl}(C_0)$; Hintikka types $\mathrm{Typ}(C_0)$ (round-12 §2);
- **Theorem A** (witness thinning; saturated split-forest presentations; found sound by all nine reviews) — used here only in the extraction direction, as a supplier of bounded-width cluster decompositions of models;
- the **RCC5 patchwork theorem** (round-12 Theorem 2.5 with Appendix A's finite tree-patching and compactness; verified in the small by WP14 and independently by the ninth review) — the single remaining external input;
- the **amalgamation-and-truth pattern** of round-14 (Theorems 3.3/4.3), certified by the ninth review; its architecture — declared data realized verbatim by patchwork-plus-compactness, truth read off declared data — is reused here with the declared data now generated by a certificate rather than guessed by run copies.

Replaced: round-14's Theorem C, its §6 monitor sketch, and with them the entire automaton layer, including the dependency on two-way alternating parity tree emptiness. Withdrawn with reasons: the ninth review's F1 shows the abstraction's enrichment cannot ride a finite tape, and independent alternating run copies cannot be forced to guess it coherently; all three natural repairs are blocked by facts the stack itself establishes. We do not attempt a fourth.

1.2 Why this is not oscillation

The automaton was adopted (round 11) because the certificate systems of rounds 2–10 could not discharge two obligations: simultaneous eventuality fulfilment, and finite exhaustion of interface data. Both obstacles have since been dissolved by results that did not exist then:

1. *Eventualities.* The parity machinery taught the project what fulfilment must mean: no request may be deferred along a repetition (the lap-collapse family of defects). In a certificate whose witnesses are *one-step* — every existential demand of every type is fulfilled inside a cluster pattern or in an adjacent one, never carried — there is nothing to defer. Infinite structure arises as infinite repetition of patterns, but each demand is discharged locally in every copy. The parity condition is not replaced; it is made unnecessary. (WP26 part C checks the historical deferral traps: $C_{3\text{hop}}$ is rejected, the towers are accepted.)

2. *Interface data.* Rounds 2–10 tried to summarize unbounded route data through interfaces; rounds 12–14 tried to guess it. The certified round-14 semantics shows what is actually needed: a single coherent global relation extending the locally declared data. Existence of such a relation is not a run-time property to be checked by a machine — it is a theorem to be proved once, from patchwork, compactness, and the composition table. The Shadow Amalgamation Lemma (round 13; found sound by reviews eight and nine) is exactly such a theorem for horizontal rows; Section 4 generalizes its method to the certificate defined here. Coherence moves from the decision device, where the ninth review proved it cannot live, into the soundness proof, where the project's certified tools live.

What remains of the decision problem after this move is finite enumeration: the certificate space is finite and its conditions are finitely checkable. No automaton, no emptiness theorem, no Vardi dependency. The cost is that two genuinely mathematical lemmas must be proved rather than delegated to a machine model; they are stated, proved to the extent the author can, and designated for attack in Section 10.

2 The fold lattice

For nonempty $A, B \subseteq \mathsf{Rel}$ let $A \circ B = \bigcup_{a\in A, b\in B} \mathrm{Comp}(a, b)$. For a word $w = a_1 \cdots a_k$ of atoms in $\mathsf{Rel} \setminus \{\mathsf{EQ}\}$ let $\mathrm{fold}(w) = \{a_1\} \circ \cdots \circ \{a_k\}$: the set of relations realizable between the endpoints of a path labelled w in some frame.

Fact 2.1 (The fold lattice; machine-checked, WP26 A1–A3)**.** Exactly ten sets arise as $\mathrm{fold}(w)$ for nonempty words w: the four singletons $\{\mathsf{PP}\}, \{\mathsf{PPI}\}, \{\mathsf{PO}\}, \{\mathsf{DR}\}$ and six non-singletons. Every non-singleton fold contains DR or PO; the three folds without DR ($\{\mathsf{PP}, \mathsf{PO}\}$, $\{\mathsf{PPI}, \mathsf{PO}\}$, $\{\mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}, \mathsf{PO}\}$) all contain PO. Consequently the *canonical selector*

$$\mathrm{sel}(F) = \begin{cases} \text{the unique element of } F \setminus \{\mathsf{EQ}\} & \text{if } |F \setminus \{\mathsf{EQ}\}| = 1, \\ \mathsf{DR} & \text{if } \mathsf{DR} \in F, \\ \mathsf{PO} & \text{otherwise} \end{cases}$$

is total on reachable folds and never selects EQ or a vertical atom unless that atom is forced.

Fact 2.2 (Steering toolkit; machine-checked, WP26 A4, WP17 A)**.** $\mathsf{DR} \in \mathrm{Comp}(W, \mathsf{DR})$ for every $W \in \mathsf{Rel}$; $\{\mathsf{DR}, \mathsf{PO}\} \subseteq \mathrm{Comp}(a, b)$ for all $a, b \in \{\mathsf{DR}, \mathsf{PO}\}$; $\mathrm{Comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ and $\mathrm{Comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$; every composition cell containing EQ, other than $\mathrm{Comp}(\mathsf{EQ}, \mathsf{EQ})$, also contains $\mathsf{PP}, \mathsf{PPI}, \mathsf{PO}$.

Fact 2.1 is the reason a finite certificate can govern unbounded distances: *along any path, the relation between two occurrences is either chain-forced (a vertical singleton, propagated by type-transitivity) or admits a horizontal value* — and horizontal values, by Fact 2.2, never constrain one another. The entire long-range analysis of a certificate happens inside a ten-element lattice.

3 Safe links and cluster quasimodels

Fix C_0 in NNF with closure $\mathrm{Cl}(C_0)$, $n = |\,\mathrm{Cl}(C_0)|$, and Hintikka types $\mathrm{Typ}(C_0)$.

Definition 3.1 (Safe links)**.** For $t, s \in \mathrm{Typ}(C_0)$ and $R \in \mathsf{Rel} \setminus \{\mathsf{EQ}\}$, the link (t, R, s) is *safe* iff

(L1) $\forall R.D \in t \Rightarrow D \in s$, and $\forall R^{-1}.D \in s \Rightarrow D \in t$;

(L2) if $R = \mathsf{PP}$: $\forall \mathsf{PP}.D \in t \Rightarrow \forall \mathsf{PP}.D \in s$; dually if $R = \mathsf{PPI}$: $\forall \mathsf{PPI}.D \in s \Rightarrow \forall \mathsf{PPI}.D \in t$.

Write $\mathrm{Safe}(t, s) = \{R : (t, R, s) \text{ safe}\}$. (L2) is the transitive propagation sound by Fact 2.2; it makes vertical chains self-policing: an obligation $\forall \mathsf{PP}.D$ holds at every element of a PP-chain above its source because the formula itself rides the types.

Definition 3.2 (Cluster pattern)**.** A *cluster pattern* over a type set $S \subseteq \mathrm{Typ}(C_0)$ is a finite structure $k = (V_k, N_k, \tau_k, \iota_k)$: a set of at most $K(C_0)$ members (the width bound of Section 6); a complete atomic composition-closed network N_k on V_k with EQ only on the diagonal; types $\tau_k : V_k \to S$ with every edge a safe link ($N_k(a, b) \in \mathrm{Safe}(\tau_k a, \tau_k b)$); and a designation ι_k of *interface* members (shared with adjacent patterns in an unfolding) and *core* members (Definition 3.3(Q5)).

Definition 3.3 (Cluster quasimodel)**.** A *cluster quasimodel* for C_0 is a tuple $Q = (S, \mathcal{K}, \mathcal{G}, \mathrm{core})$: a type set $S \subseteq \mathrm{Typ}(C_0)$; a finite catalogue $\mathcal{K}$ of cluster patterns over S; a *gluing graph* $\mathcal{G}$ whose edges identify an interface of one pattern with an interface of another (equal types, equal networks and safe links on the shared part); and a distinguished *core* pattern. Conditions:

(Q1) **Root.** Some pattern has a member with C_0 in its type.

(Q2) **One-step demands.** For every pattern k, every *interior* (non-interface) member a, and every $\exists R.D \in \tau_k(a)$: either some $b \in V_k$ has $N_k(a, b) = R$, $D \in \tau_k(b)$ (in-pattern fulfilment), or a is an interface member of a $\mathcal{G}$-adjacent pattern in which the demand is fulfilled in-pattern. No demand is ever carried further than one gluing step; there are no promissory witnesses, requests, or fulfilment declarations.

(Q3) **Gluing coherence.** Adjacent patterns agree on their shared interface (types, network, links), so that every finite connected unfolding is a tree-shaped agreeing family of finite closed atomic networks.

(Q4) **Fold conditions.** Let $R(k, k') \subseteq$ fold lattice be the (finitely computable) set of folds of separator-mediated paths between occurrences of patterns k and k' in unfoldings of $\mathcal{G}$ (a fixed point over the ten-element lattice on the gluing graph). For all patterns k, k', members $a \in V_k$, $c \in V_{k'}$, and every $F \in R(k, k')$:

- (a) if $F \setminus \{\mathsf{EQ}\}$ is a singleton $\{V\}$ (a chain-forced value), then $V \in \mathrm{Safe}(\tau_k a, \tau_{k'} c)$;
- (b) otherwise $\mathrm{sel}(F) \in \mathrm{Safe}(\tau_k a, \tau_{k'} c)$ or the other horizontal member of F (if any) is in $\mathrm{Safe}(\tau_k a, \tau_{k'} c)$.

(Q5) **Unique types and the core.** Call $t \in S$ *unique* if $\mathrm{Safe}(t, t) \cap (\mathsf{Rel} \setminus \{\mathsf{EQ}\}) = \varnothing$ (two distinct realizations of t could stand in no relation at all). Every member whose type is unique is a *core member*; the core pattern contains all core members with a fixed closed network among them; every other pattern contains the core members it references as permanent interface members, glued toward the core through the unfolding (so a unique type has exactly one realization, shared globally — exact occurrence identity, not equality-quotienting).

(Q6) **Width.** $|V_k| \le K(C_0)$ for every k, with $K(C_0)$ the bound of Lemma 6.2 (which includes the core and one-step witness overhead).

Remark 3.4 (What replaced what)**.** (Q2) replaces witness/request declarations and the parity condition. (Q4) replaces relation-vector shadows, interface transformers, and the active enrichment: nothing is declared for distant pairs at all — (Q4) merely ensures that, at realization time, the canonical choice on the fold lattice is always safe. (Q5) replaces the equality-port / typed-EQ machinery for semantically unique elements and absorbs the ninth review's probe p4 structurally: the globally unique witness sits in the core, every tower cluster references it through its interface, and no source is ever dragged along a ray.

4 Soundness: from certificate to model

Throughout, fix a cluster quasimodel Q for C_0.

Definition 4.1 (Unfolding)**.** The *unfolding* $U(Q)$ is the countable tree of pattern copies generated from a root copy of a (Q1)-pattern by satisfying (Q2): for every interior demand delegated to an adjacent pattern, attach a fresh copy of that pattern glued on the shared interface; core members are never copied — every reference to a core member is identified with the single core instance, and the core pattern is attached at the root. By (Q3) and (Q5), $U(Q)$ is a tree-shaped family of finite closed atomic networks agreeing on overlaps, in which every occurrence's trace is connected (interfaces and core references are shared along tree paths; running intersection holds by construction).

Lemma 4.2 (Skeleton realization)**.** *There is a partial atomic relation ρ_0 on the occurrences of $U(Q)$, defined exactly on co-patterned pairs, extending every pattern network, such that every finite connected sub-family is a closed atomic network: namely the union of the pattern networks, which is coherent by (Q3).*

Proof. Immediate from (Q3) and connectedness of traces: two co-patterned occurrences receive the same value in every shared pattern. ☐

Lemma 4.3 (Canonical completion)**.** *ρ_0 extends to a total complete atomic composition-closed relation ρ on the occurrences of $U(Q)$, with EQ only on the diagonal, such that for every non-co-patterned pair (x, y), $\rho(x, y) \in \mathrm{Safe}(\mathrm{tp}(x), \mathrm{tp}(y))$.*

Proof. We construct ρ over an exhausting chain $U_1 \subseteq U_2 \subseteq \cdots$ of finite connected sub-unfoldings (each U_{m+1} adds one pattern copy, glued on its separator σ_m; the core is inside U_1). Invariant: ρ is total, closed, atomic on U_m's occurrences, extends ρ_0 there, and all non-co-patterned values are safe.

Base: U_1 is a single pattern copy plus the core, glued; finite patchwork (two agreeing closed networks over the shared core interface) gives a closed atomic completion; its new values are separator-mediated, so each lies in the fold of its connecting path; by (Q4) a safe choice exists in that fold, and we must show the *joint* choice can be made safely — this is the same argument as the inductive step, below, with U_0 the core.

Step: let A be the realized structure on U_m (a finite closed atomic network, hence — by the patchwork property — a globally consistent one), and let k be the new pattern copy with separator $\sigma \subseteq A$. For each fresh member b of k and each old occurrence y, the set of values v such that $A \cup k \cup \{(y, b) \mapsto v\}$ admits a closed completion is nonempty (two-bag patchwork on A and k over σ) and is contained in the fold $F(y, b)$ of the tree path from y through σ to b (composition closure);

conversely every value in the pointwise minimal completion set is realizable by global consistency of A. Assign:

$$\rho(y,b) := \begin{cases} \text{the forced value} & \text{if } F(y,b) \setminus \{\mathsf{EQ}\} \text{ is a singleton,} \\ \text{a horizontal member of } F(y,b) \cap \mathrm{Safe}(\mathrm{tp}\, y, \mathrm{tp}\, b) & \text{otherwise,} \end{cases}$$

where in the second case a horizontal member exists by Fact 2.1 and is safe by (Q4); when both horizontals are available and safe we fix $\mathrm{sel}(F(y,b))$. Safety in the first case is (Q4)(a). It remains to check that the assignment is jointly closed.

Triangles with at most one fresh member and two old members reduce to the invariant plus the definition of F (the assigned value lies in every composition through old occurrences, because $F(y,b)$ is computed along the tree path and A is globally consistent: any old detour composes to a superset of the tree-path fold). Triangles with two fresh members (y,b,b'): $\rho(b,b')$ is the pattern value; $F(y,b) \subseteq F(y,b') \circ \{N_k(b',b)\}$ and the forced/horizontal cases match by the fold algebra: if either side is forced the other's constraint set contains the assigned value by definition of fold; if both are free, both assigned values are horizontal, and the connecting constraint $\rho(y,b) \in \mathrm{Comp}(\rho(y,b'), N_k(b',b))$ holds because the fold $F(y,b)$ was computed through the same separator and the horizontal choice was taken *inside* $F(y,b)$. Triangles among three old members hold by the invariant. Finally, atomicity: no assigned value is EQ (the selector never selects EQ; forced EQ is impossible by Fact 2.2 and distinctness of occurrences).

The limit of the chain is total on $\bigcup_m U_m = U(Q)$, closed (every triple lies in some U_m), atomic, and safe off-pattern. □

Remark 4.4 (Honesty about Lemma 4.3)**.** The two-fresh-member and old-detour steps above compress a fold-algebra argument whose fully expanded form is the designated weak point W1 (Section 10). Its empirical standing: WP26 part B realizes 250/250 random glue steps with canonical-selector domains and sees 245/249 failures when the horizontal choices are stripped (the F1.3/WP15 genus the conditions exclude); WP17 parts B/C support the same discipline at the row level; reviews eight and nine both re-ran the corresponding checks and found no defect in the horizontal-steering core. What is genuinely new relative to round 13's certified lemma is the interleaving of forced vertical values with steered horizontal ones along tree paths — exactly what (Q4)(a) and the fold lattice were introduced to govern.

Theorem 4.5 (Soundness)**.** *If a cluster quasimodel for C_0 exists, then C_0 is satisfiable in an abstract complete atomic RCC5 frame with strong equality.*

Proof. Take ρ from Lemma 4.3 on the occurrences of $U(Q)$; interpret concept names by types. (Δ, ρ) is a frame: (R1) by atomicity and diagonal EQ; (R2), (R3) by closure. Truth lemma, by induction on $C \in \mathrm{Cl}(C_0)$: atomic and Boolean cases from the type conditions. *Existential:* $\exists R.D \in \mathrm{tp}(x)$ is fulfilled in-pattern or one step away (Q2); the witness edge is a pattern value, extended verbatim by ρ; induction applies. *Universal:* let $\forall R.D \in \mathrm{tp}(x)$ and $\rho(x,y) = R$. If x,y are co-patterned, the pattern edge is a safe link (Definition 3.2), so $D \in \mathrm{tp}(y)$ by (L1). Otherwise $\rho(x,y)$ was assigned by Lemma 4.3 *inside* $\mathrm{Safe}(\mathrm{tp}\, x, \mathrm{tp}\, y)$ — whether forced (Q4)(a) or steered (Q4)(b) — so again (L1) gives $D \in \mathrm{tp}(y)$. There is no third case: every pair is co-patterned or Lemma-4.3-assigned. The contrapositive direction (type completeness supplies $\exists R.\overline{D}$) is as in round 14. The root (Q1) gives C_0 at its occurrence. □

Remark 4.6. Compare the corresponding round-14 step: there, the universal case needed the run-time device to have guessed the very value $\rho(x,y)$ coherently — the obligation the ninth review proved undischargeable. Here the value is chosen *in the proof*, from a safe set whose nonemptiness is a finite certificate condition. No guessing, no monitors, no tape.

5 Completeness: from model to certificate

Theorem 5.1 (Extraction)**.** *If C_0 is satisfiable, a cluster quasimodel for C_0 exists.*

Proof. Let $\mathcal{M} \models C_0$ at a_0; apply Theorem A: witness thinning and saturation give a witness-generated presentation of width $K(C_0)$ (Lemma 6.2) whose bags are finite closed atomic networks (restrictions of $\mathcal{M}$) fulfilling each selected demand within the bag of its source or the adjacent bag that introduces the witness — the presentation's bags *are* cluster patterns, with types read off $\mathcal{M}$ and every edge safe because $\mathcal{M}$ satisfies its types. Set S = realized types; $\mathcal{K}$ = the finitely many isomorphism classes of decorated bags (finite by Lemma 6.2); $\mathcal{G}$ = the realized adjacencies. (Q1)–(Q3) hold by construction. (Q5): a unique type has at most one realization in $\mathcal{M}$ (any two distinct realizations would stand in some atomic relation, contradicting $\mathrm{Safe}(t,t) \setminus \{\mathsf{EQ}\} = \varnothing$ since $\mathcal{M}$ satisfies its types); place these realizations in the core bag — there are at most $|S|$ of them — and thread the core through the presentation as permanent interface members (width accounted in Lemma 6.2). (Q4): for a fold $F \in R(k,k')$ realized between occurrences a,c in $\mathcal{M}$, the true relation $\rho_{\mathcal{M}}(a,c)$ lies in F and is safe; for singleton folds this is exactly (Q4)(a). For non-singleton folds, (Q4)(b) requires a *horizontal* safe value, and the true relation need not be horizontal; here extraction uses the model's freedom: by global consistency of $\mathcal{M}$'s finite restrictions and the fold fact, either some placement of the two types at that fold distance realizes a horizontal relation — which is then safe, since realized — or every model of these types at this fold distance realizes a vertical relation, in which case the pair is chain-confined and the fixed-point computation of $R(k,k')$ yields the singleton sub-fold along which (Q4)(a) applies. This dichotomy is the designated weak point W2: its full proof requires a normal-form argument (every non-chain-forced coexistence can be re-presented horizontally) that is stated here as Lemma 5.2 and proved only in outline. $\square$

Lemma 5.2 (Horizontal normal form; designated W2)**.** *Let t, s be realized types of $\mathcal{M}$ and suppose some presentation of $\mathcal{M}$ places occurrences of t and s at a non-singleton fold F. Then there is a presentation in which their relation at that placement is a horizontal member of F, or a refinement of the gluing graph on which the fixed point $R(k,k')$ excludes F for this pair.* Proof outline: *a non-singleton fold leaves the pair's value undetermined by the path; by global consistency of the presentation's finite restrictions the horizontal member of F is realizable at the pair unless a* different *path forces verticality, in which case that path's singleton fold governs and the fixed point records it. The interaction of several forcing paths is where care is needed; Fact 2.1 bounds the interaction to the ten-set lattice.*

6 The width bound, re-proved (resolves F3)

The ninth review's F3 found the previous width lemma's recurrence unclosed — its witness-generation reason (g2) could cascade at fixed rank — and its provenance contradictory. We re-prove the bound from a single decreasing measure and re-house it in this manuscript.

Definition 6.1 (Generation measure)**.** Every member of a cluster pattern carries a *generation reason*: (g1) the source of the pattern; (g2) a selected witness of a demand $\exists R.D$ of an already-generated member a, with measure $\mu(b) :=$ the modal depth of D; (g3) a boundary/support insertion for an already-generated member a, with $\mu(b) := \mu(a)$ (there are at most $c_3(n)$ such insertions per member, a fixed count of parents, selected supports, and interface manifestations); (g4) a stratum/threshold insertion for a pair, with $\mu(b) :=$ min of the pair's measures; (g5) a core reference (no new member: core members are globally shared and counted once). The root's measure is the modal depth of C_0.

Lemma 6.2 (Closed recurrence)**.** *Under (g2) the measure strictly decreases: the witness of $\exists R.D$ receives measure* $\mathrm{md}(D) < \mathrm{md}(\exists R.D) \le \mu(a)$*. (g3) and (g4) do not increase the measure and are applied at most a fixed number of times per member and per pair respectively, with no iteration (a boundary insertion for an insertion is itself charged to the original member's budget $c_3(n)$; a stratum for a pair involving a stratum is charged to the original pair). Hence, writing W_d for the maximal number of members of measure $\ge d$ generated from one source,*

$$W_d \;\le\; \big(1 + c_3(n)\big)\,\big(1 + n\,W_{d+1}\big) + \; c_4(n)\,\big((1 + c_3(n))(1 + n\,W_{d+1})\big)^2, \qquad W_{\mathrm{md}(C_0)+1} = 0,$$

and $K(C_0) := W_0 + |\,\mathrm{Typ}(C_0)|$ (the second summand covers the core) is computable.

Proof. The only unbounded-iteration risk in the previous version was (g2)-at-fixed-rank; with μ decreasing strictly at every (g2) step that risk is gone: a chain of generations contains at most $\mathrm{md}(C_0)$ many (g2) steps, and between consecutive (g2) steps only the fixed budgets c_3, c_4 apply, by definition of the reasons. The displayed recurrence just counts: per member, its (g3) budget; per member, its at most n demands each opening a strictly-smaller-measure subtree; per pair of the resulting members, its (g4) budget. Every member of a saturated pattern has a reason (that is what witness-generated means), so the pattern is contained in the counted set. □

Remark 6.3. No optimality is claimed; decidability needs computability only. The request-source liveness pressure of the ninth review's F2 does not arise in this architecture: there are no requests to keep alive — (Q2) fulfils every demand within one gluing step, and the shared-witness towers of probe p4 are carried by the core (one permanent interface member), not by per-source liveness. The residual width obligation is (W3): that Theorem A's saturation can always be arranged to fulfil demands one-step at width $K(C_0)$ — the extraction in Theorem 5.1 inherits exactly the saturation apparatus all nine reviews have examined, now with the explicit measure above.

7 Decidability

Theorem 7.1 (Decidability)**.** *Concept satisfiability for $\mathcal{ALCI}_{\mathrm{RCC5}}$ over abstract complete atomic RCC5 frames with strong equality is decidable, assuming the RCC5 patchwork property.*

Proof. The certificate space is finite: S ranges over subsets of $\mathrm{Typ}(C_0)$; patterns are decorated networks on at most $K(C_0)$ members over finitely many types and atoms; catalogues and gluing graphs over finitely many patterns; cores over finitely many members. Conditions (Q1)–(Q6) are finitely checkable — (Q4)'s fold sets are a fixed point over a ten-element lattice on the finite gluing graph (Fact 2.1); everything else is a table check. By Theorems 4.5 and 5.1, C_0 is satisfiable iff a certificate exists. Enumerate. □

No complexity claim is made. The practical procedure remains the cover-tree tableau, which decides all diagnostic concepts of all nine reviews in agreement with this semantics (WP26 part C: eleven diagnostics and a 200-concept random sweep, zero disagreements).

8 Machine checks (WP26)

`verification/python/wp26_round15_cluster_quasimodels.py`, all PASS: **A** the fold lattice has exactly ten reachable sets; the FOLD fact; the selector's totality; the steering toolkit — exhaustively, against a table re-derived from set semantics. **B** the steering step of Lemma 4.3 on 250 random realized cluster trees with a fresh glued pattern: canonical-selector domains jointly realizable

250/250; stripping the horizontal choices produces 245/249 failures (the F1.3/WP15 genus that (Q4) excludes). **C** acceptance cross-validation against the cover-tree tableau: C_{force}, C_{split}, C_{2hop}, C_{3hop}, a sanity concept UNSAT; $C_{\text{recursive}}$, $C_{\uparrow}$, PO-loop, dual-descendant, *the eighth review's* C_{G2a}, and *the ninth review's* p4 shared-unique-witness tower SAT; 200 random concepts, zero disagreements, zero timeouts.

9 Response map to the ninth review

R1/R2	Taken in the R2 alternative: the automaton layer is removed; the decision notion is the finite cluster-quasimodel certificate; the enrichment is never guessed — it is constructed in Lemma 4.3. The F1.3 acceptance-without-validity test is structurally moot (there are no run copies), and its configuration is excluded at realization time by (Q4) + Fact 2.1 (WP26 B negative control).
R3	The request-delay problem is dissolved rather than bounded: (Q2) admits no deferral, and probe p4's shared witness is a core member (one permanent interface occurrence), not a liveness cost per source.
R4	Section 6: single strictly decreasing measure (remaining modal depth at (g2)); fixed budgets elsewhere; closed recurrence; the width proof now lives in the same document that uses it.
R5	This manuscript states its full certificate language (Sections 3–6) without deltas; it inherits only Theorem A, the patchwork theorem, and the certified round-14 semantic pattern, each by explicit citation. A single consolidated edition merging the Theorem A material remains editorial work.
R6	WP26 ships with this manuscript; WP14/WP17 remain the checks for the inherited facts.

10 What would invalidate this manuscript

(W1) **The steering induction (Lemma 4.3).** The two-fresh-member and old-detour triangle cases are proved by a compressed fold-algebra argument. A configuration in which the canonical assignment violates closure — necessarily involving an interleaving of chain-forced vertical values and steered horizontal ones that the ten-set lattice analysis misses — would break soundness. WP26 B searched (250 random glue steps, 0 failures; negative control fires), but the induction deserves a fully expanded case table or mechanization; this is the manuscript's keystone.

(W2) **The horizontal normal form (Lemma 5.2).** Extraction needs every non-chain-forced coexistence to be presentable horizontally. A satisfiable concept two of whose types can co-occur *only* vertically, at a placement whose fold the fixed point fails to classify as chain-forced, would break completeness (an over-rejection). The interaction of multiple forcing paths is the suspect step.

(W3) **One-step saturation at width $K(C_0)$.** Theorem A must supply presentations whose every demand is fulfilled within one gluing step at the re-proved width. This inherits the saturation apparatus (S1)–(S6) all nine reviews have examined; the new exposure is the combination with the core threading of (Q5).

(W4) **The external input.** The RCC5 patchwork property, as before (WP14; ninth review S9) — now the *only* external theorem: the two-way parity emptiness dependency is gone with the automaton.

11 Status

The decidability theorem for $\mathcal{ALCI}_{\mathrm{RCC5}}$ remains **strongly supported, not certified**. This manuscript is the tenth attempt at the decision layer; nine reviews have found nine defects, and the reader should presume this round contains the tenth until an adversarial review fails to find it. What can be said structurally: the semantic half is inherited from a cold-certified base; the decision half no longer contains a guessing device to be unsound about — its two load-bearing lemmas (W1, W2) are ordinary finite combinatorics over a ten-element lattice, stated with their machine evidence, and are exactly the right size for either an adversarial review or a mechanization to settle.

References

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[3] Fresh Fable 5 session. *Cold Referee Report: Decidability of ALCI_RCC5 Concept Satisfiability* (packets 1–3), July 9, 2026. `papers/cold_review_round14/`.

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